

Week 1

1. What kind of software and hardware will be on the drone?

- Wifi antenna
- GPS receiver/ other sensors
- Cube (flight controller)
- NVIDIA Jetson Orin Nano
- 360 auto-stabilize gimbal camera

- Full Software & Hardware Spec List
 - Object Detection (Week 3)
 - Historically done with OpenCV
 - Aerial Imagery/ Stitching (Week 2)
 - Mapping and surveying
 - Fly over predefined area, find objects or map
 - Flight Controller (Week 4)
 - Hardware → Cube Orange+, Pixhawk
 - Firmware → ArduPilot, Betaflight, PX4
 - Keeps drone stable and flying
 - Takes commands via MAVLink over a physical link (UART serial cable) from the onboard computer.
 - Onboard computer, sensors, & middleware (Week 4,5, and 6)
 - Jetson/Pi running Ubuntu → main brain of system
 - Runs on code written under the ROS framework that tells the flight controller what to do
 - The code takes the flight controller sensor data as input, processes this data using the logics/algorithms you will implement, and generates fly commands.
 - ROS helps get data from and send commands to the hardware.
 - ROS (week 5 & 6)

2. Projects

- Priority → competitions
 - Revolve around autonomy missions:
 - Time-based waypoint flying
 - Mapping/surveying
 - Object detection, classification, and localization
 - Package drop/payload delivery at targets
- OpenDroneMap
- Basic Autonomy and Production Autonomy
- Industry pushing in BVLOS: Beyond Visual Line of Sight and multiple drones (FAA Part 108)

Week 2:

NADIR imaging - lower oblique angles

Gimbal camera

How is OpenCV used

OpenCV: General-purpose computer vision library, does almost everything

- Feature detection
- 3d reconstruction
- Motion analysis
- Optical flow
- Photo manipulation/refining
- Video editing/read/write

We are using: Feature detection (SIFT, ORB), featuring matching, image manipulation.

OpenDroneMap (ODM) workflow

- Get pictures with Geo tags
- Adjust settings

ODM benefits → a lot simpler to use, can sorta avoid coding

Drawbacks → not optimized for being done in flight, can be slow unless carefully optimized with

settings → requires accurate GPS tags, run poorly without GPS

This year

→ make a custom image stitching/SFM (structure from motion) algorithm with OpenCV that can run real-time on the drone

→ Dive deep into OpenDroneMap specifics and figure out how to neatly divide up the stitching up.

Other potential projects

- SLAM
- Optical Flow
- Structure from Motion
- Applying drone imagery